Rational Expressions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Consider the quadratic function $f(x) = x^2 - 6x + 8$. What are the roots of the equation $f(x) = 0$?

To find the roots of the equation $f(x) = 0$, we need to solve $x^2 - 6x + 8 = 0$. This quadratic equation can be factored. We are looking for two numbers that multiply to 8 and add up to -6. These numbers are -2 and -4. Therefore, the equation can be factored as $(x - 2)(x - 4) = 0$. Setting each factor to zero gives us the roots: $x - 2 = 0 \Rightarrow x = 2$ and $x - 4 = 0 \Rightarrow x = 4$.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Consider the polynomial $P(x) = x^3 - 6x^2 + kx + 12$. If $(x - 3)$ is a factor of $P(x)$, and the sum of the roots of $P(x) = 0$ is S, find the value of k. Additionally, determine the sum of the roots S given that the roots are α, β, γ.

Since $(x - 3)$ is a factor of $P(x) = x^3 - 6x^2 + kx + 12$, we know that $P(3) = 0$. Substituting $x = 3$ into the polynomial: $P(3) = (3)^3 - 6(3)^2 + k(3) + 12 = 0$ $27 - 6(9) + 3k + 12 = 0$ $27 - 54 + 3k + 12 = 0$ $-27 + 3k + 12 = 0$ $-15 + 3k = 0$ $3k = 15$ $k = 5$. For a cubic polynomial of the form $ax^3 + bx^2 + cx + d = 0$, the sum of the roots $(\alpha + \beta + \gamma)$ is given by $-\frac{b}{a}$. In our case, $P(x) = x^3 - 6x^2 + kx + 12$, so $a = 1$ and $b = -6$. The sum of the roots.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The rational expression $\frac{2x^2 + ax - 7}{x + 4}$ is equivalent to $2x + b + \frac{5}{x + 4}$ for all $xeq -4$, where a and b are constants. What is the value of $a + b$?

To find the values of a and b, equate the two expressions: $\frac{2x^2 + ax - 7}{x + 4} = 2x + b + \frac{5}{x + 4}$ Multiplying both sides by the common denominator $x + 4$: $2x^2 + ax - 7 = (2x + b)(x + 4) + 5$ Expand the right side: $(2x + b)(x + 4) + 5 = 2x^2 + 8x + bx + 4b + 5 = 2x^2 + (b + 8)x + (4b + 5)$ Equating coefficients of corresponding powers of x on both sides: 1. Constant terms: $-7 = 4b + 5 \Rightarrow 4b = -12 \Rightarrow b = -3$. 2. Linear coefficients: $a = b + 8 \Rightarrow a = -3 + 8 = 5$.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The quadratic function f is defined by $f(x) = (x + 3)^2 - 5$. What are the coordinates of the vertex of the graph of $y = f(x)$ in the xy-plane?

The vertex form of a quadratic function is $f(x) = a(x - h)^2 + k$, where the point (h, k) represents the vertex of the parabola. Rewriting $f(x) = (x + 3)^2 - 5$ in this standard form: $f(x) = (x - (-3))^2 + (-5)$ From this expression, we identify: - $h = -3$ - $k = -5$ Thus, the coordinates of the vertex are $(-3, -5)$. Common Misconceptions: - $(3, -5)$: Incorrectly using $+3$ directly from the expression without accounting for the negative sign in $(x - h)$. - $(-3, 5)$ or $(3, 5)$: Inverting the sign of the.

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The quadratic function $f(x) = ax^2 + bx + c$ has a vertex at the point $(2, -5)$. If the graph of $f(x)$ passes through the point $(4, 3)$, find the value of a.

The vertex form of a quadratic function is $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. Given the vertex is $(2, -5)$, the function can be written as $f(x) = a(x - 2)^2 - 5$. We are also given that the graph passes through the point $(4, 3)$. Substituting these coordinates into the equation, we get: $3 = a(4 - 2)^2 - 5$. Simplifying, we have $3 = a(2)^2 - 5$, which means $3 = 4a - 5$. Adding 5 to both sides gives $8 = 4a$. Dividing by 4, we find $a = 2$.

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Factor the quadratic expression: $x^2 + 5x + 6$.

To factor the quadratic expression $x^2 + 5x + 6$, we need to find two numbers that multiply to 6 and add up to 5. These two numbers are 2 and 3. Therefore, the factored form is $(x + 2)(x + 3)$. Let's check the other options: - $(x + 1)(x + 6) = x^2 + 6x + x + 6 = x^2 + 7x + 6$ (Incorrect sum) - $(x - 2)(x - 3) = x^2 - 3x - 2x + 6 = x^2 - 5x + 6$ (Incorrect sum sign) - $(x - 1)(x - 6) = x^2 - 6x - x + 6 = x^2 - 7x + 6$ (Incorrect sum sign)

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the rational expression: $\frac{x^2 - 4}{x - 2}$. If $xeq2$, what is the simplified form of this expression?

The given rational expression is $\frac{x^2 - 4}{x - 2}$. The numerator $x^2 - 4$ is a difference of squares, which can be factored as $(x - 2)(x + 2)$. Therefore, the expression can be rewritten as $\frac{(x - 2)(x + 2)}{x - 2}$. Since it is given that $xeq2$, we can cancel the common factor $(x - 2)$ from the numerator and the denominator. This leaves us with $x + 2$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Given the polynomial $P(x) = ax^3 - 2x^2 + bx + 5$, if $P(x)$ is divisible by $(x - 1)$ and the remainder when $P(x)$ is divided by $(x + 2)$ is -9, find the value of $a + b$.

Since $P(x)$ is divisible by $(x - 1)$, by the Remainder Theorem, $P(1) = 0$. Substituting $x = 1$ into $P(x)$: $a(1)^3 - 2(1)^2 + b(1) + 5 = 0 \Rightarrow a - 2 + b + 5 = 0 \Rightarrow a + b + 3 = 0 \Rightarrow a + b = -3$. When $P(x)$ is divided by $(x + 2)$, the remainder is -9. By the Remainder Theorem, $P(-2) = -9$. Substituting $x = -2$ into $P(x)$: $a(-2)^3 - 2(-2)^2 + b(-2) + 5 = -9$. $-8a - 2(4) - 2b + 5 = -9$ $-8a - 8 - 2b + 5 = -9$ $-8a - 2b - 3 = -9$ $-8a - 2b = -6$ Dividing by -2: $4a + b = 3$. We have a s.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

If the expression $\frac{\sqrt[4]{x^3 \cdot \sqrt{x^k}}}{x^{-\frac{1}{4}}}$ is equivalent to x^2 for all $x > 0$, what is the value of the constant k?

To find the value of k, convert each radical expression into rational exponent form and apply exponent rules step-by-step: 1. Convert the inner radical: $\sqrt{x^k} = (x^k)^{\frac{1}{2}} = x^{\frac{k}{2}}$ 2. Combine terms inside the fourth root using the product rule $x^a \cdot x^b = x^{a+b}$: $x^3 \cdot x^{\frac{k}{2}} = x^{3+\frac{k}{2}} = x^{\frac{6+k}{2}}$ 3. Apply the fourth root using the power rule $(x^a)^b = x^{ab}$.

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Triangle ABC is mapped to triangle $A'B'C'$ on the coordinate plane as shown below. Which coordinate rule represents this transformation?

To determine the transformation rule, compare the coordinates of corresponding vertices of triangle ABC and triangle $A'B'C'$: - Vertex $A(1, 2)$ maps to $A'(-4, 2)$. - Vertex $B(4, 2)$ maps to $B'(-1, 2)$. - Vertex $C(2, 5)$ maps to $C'(-3, 5)$. For each vertex, the change in the x-coordinate is: $\Delta x = -4 - 1 = -5$ The y-coordinate remains unchanged: $\Delta y = 2 - 2 = 0$ This is a horizontal translation of 5 units to the left, which is represented by the transformation rule.

11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Let $f(x) = \sqrt{x^2 - 4x + 4}$ and $g(x) = \sqrt[3]{x^3 + 6x^2 + 12x + 8}$. If $h(x) = f(x) \cdot g(x)$, find the value of $h(-3)$.

First, simplify the expressions for $f(x)$ and $g(x)$. For $f(x) = \sqrt{x^2 - 4x + 4}$, we recognize the expression inside the square root as a perfect square trinomial: $x^2 - 4x + 4 = (x - 2)^2$. Therefore, $f(x) = \sqrt{(x - 2)^2} = |x - 2|$. For $g(x) = \sqrt[3]{x^3 + 6x^2 + 12x + 8}$, we recognize the expression inside the cube root as a perfect cube: $x^3 + 6x^2 + 12x + 8 = (x + 2)^3$. Therefore, $g(x) = \sqrt[3]{(x + 2)^3} = x + 2$. Now, we can find the expression for $h(x) = f(x) \cdot g(x)$..

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12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the function defined by $f(x) = \sqrt{ax + b}$. If the graph of $f(x)$ passes through the points $(1, 3)$ and $(5, 5)$, determine the value of a.

The function is given by $f(x) = \sqrt{ax + b}$. The graph passes through $(1, 3)$, so $f(1) = 3$. Substituting these values into the function: $3 = \sqrt{a(1) + b}$. Squaring both sides gives $9 = a + b$ (Equation 1). The graph also passes through $(5, 5)$, so $f(5) = 5$. Substituting these values into the function: $5 = \sqrt{a(5) + b}$. Squaring both sides gives $25 = 5a + b$ (Equation 2). We now have a system of two linear equations with two variables: $a + b = 9$ and $5a + b = 25$. Subtract Equation 1 from Equ.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In the figure below, triangle ABC is a right triangle with right angle at C, and altitude CD is drawn to hypotenuse AB. The length of altitude CD is 6, $AD = x$, and $DB = x + 5$. What is the length of hypotenuse AB?

By the Geometric Mean Theorem (Altitude Rule) for right triangles, the altitude to the hypotenuse is the geometric mean of the two segments of the hypotenuse: $CD^2 = AD \cdot DB$ Substitute the given values into the equation: $6^2 = x(x + 5)$ $36 = x^2 + 5x$ Rearrange into standard quadratic form: $x^2 + 5x - 36 = 0$ Factor the quadratic equation: $(x + 9)(x - 4) = 0$ This gives two solutions: $x = -9$ and $x = 4$. Because $AD = x$ represents a geometric length, it must be positive. Thus, we reject the extraneous.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the function $y = a \cdot b^x$, where $ae0$ and $b > 0, be1$, passes through the points $(1, 6)$ and $(3, 54)$. Determine the exact value of a and the exact value of b.

Let the function be $y = a \cdot b^x$. Since the graph passes through $(1, 6)$, we have $6 = a \cdot b^1$ (Equation 1). Since the graph also passes through $(3, 54)$, we have $54 = a \cdot b^3$ (Equation 2). From Equation 1, we can express a in terms of b: $a = \frac{6}{b}$. Substitute this expression for a into Equation 2: $54 = \left(\frac{6}{b}\right) \cdot b^3$ $54 = 6 \cdot b^2$ Divide both sides by 6: $b^2 = \frac{54}{6}$ $b^2 = 9$ Since $b > 0$, we take the positive square root: $b = 3$. Now substitute.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

The graph of the function $f(x) = a \cdot b^x$ passes through the points $(0, 3)$ and $(2, 75)$. Determine the value of a.

The function is given by $f(x) = a \cdot b^x$. \\ We are given that the graph passes through $(0, 3)$. Substituting these coordinates into the function: \\ $3 = a \cdot b^0$\\ Since $b^0 = 1$ (for any non-zero b), we have: \\ $3 = a \cdot 1$\\ $a = 3$. \\ Now, we can use the second point $(2, 75)$ to find b. \\ $75 = a \cdot b^2$\\ Substitute the value of $a = 3$: \\ $75 = 3 \cdot b^2$\\ Divide both sides by 3: \\ $25 = b^2$\\ Taking the square root of both sides, we get $b = 5$ (assuming $b > 0$ for typical expo.

16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Systems & Inequalities

The graph of the function $f(x) = \sqrt{ax + b}$ passes through the points $(1, 3)$ and $(5, 5)$. Find the value of a and the value of b.

Given the function $f(x) = \sqrt{ax + b}$, and it passes through the points $(1, 3)$ and $(5, 5)$. We can set up a system of equations: 1. For point $(1, 3)$: $3 = \sqrt{a(1) + b} \Rightarrow 3 = \sqrt{a + b}$. Squaring both sides gives $9 = a + b$. (Equation 1) 2. For point $(5, 5)$: $5 = \sqrt{a(5) + b} \Rightarrow 5 = \sqrt{5a + b}$. Squaring both sides gives $25 = 5a + b$. (Equation 2) Now we solve the system of linear equations: Equation 2 - Equation 1: $(5a + b) - (a + b) = 25 - 9$ $4a = 16$ $a = 4$ Substitute the value of a.

17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Sequences & Series

The graph of the exponential function $f(x) = a \cdot b^x + c$ passes through the points $(0, 3)$ and $(1, 7)$. If the horizontal asymptote of the graph is $y = 1$, find the value of a and b.

The general form of the exponential function is $f(x) = a \cdot b^x + c$. We are given that the horizontal asymptote is $y = 1$, which means $c = 1$. So, the function becomes $f(x) = a \cdot b^x + 1$. ; The graph passes through $(0, 3)$. Substituting these coordinates into the function: ; $3 = a \cdot b^0 + 1; 3 = a \cdot 1 + 1; 3 = a + 1; a = 2$. ; Now, the function is $f(x) = 2 \cdot b^x + 1$. ; The graph also passes through $(1, 7)$. Substituting these coordinates into the function.

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Factoring Polynomials

Let $P(x)$ be a polynomial such that $P(x^2 + 1) = x^4 + 3x^2 + 2$. If $P(x)$ can be expressed in the form $ax^2 + bx + c$, find the value of b.

We are given the equation $P(x^2 + 1) = x^4 + 3x^2 + 2$. We are also told that $P(x)$ is a quadratic polynomial of the form $ax^2 + bx + c$. Let $y = x^2 + 1$. Then $x^2 = y - 1$. Substituting this into the expression for $P(x^2 + 1)$, we get: $P(y) = (y - 1)^2 + 3(y - 1) + 2$ $P(y) = (y^2 - 2y + 1) + (3y - 3) + 2$ $P(y) = y^2 + (-2y + 3y) + (1 - 3 + 2)$ $P(y) = y^2 + y + 0$ Thus, $P(y) = y^2 + y$. Replacing y with x, we have $P(x) = x^2 + x$. Comparing this to the form $P(x) = ax^2 + bx + c$, we find that $a = 1$, $b = 1$, an.

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19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 9.01 Radicals & Exponents

Simplify the complex rational expression: $\frac{\frac{1}{x}-\frac{1}{y}}{\frac{1}{x^2}-\frac{1}{y^2}}$, where $xeqy$ and $xyeq0$.

Factor the denominator as a difference of squares: $\frac{1}{x^2}-\frac{1}{y^2}=\left(\frac{1}{x}-\frac{1}{y}\right)\left(\frac{1}{x}+\frac{1}{y}\right)$. Cancel the common factor $\left(\frac{1}{x}-\frac{1}{y}\right)$: $\frac{1}{\frac{1}{x}+\frac{1}{y}}=\frac{1}{\frac{y+x}{xy}}=\frac{xy}{x+y}$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Linear Equations & Lines

[Rational Expressions · Practice 3] If the governing relation for this system is $g(t)=5t+5$, evaluate $g(8)$ and find the inverse input t when $g(t)=50$.

Substitute $t=8$: $g(8)=5(8)+5=\mathbf{45}$. Setting $5t+5=50$ gives $5t=45\Rightarrow t=\mathbf{9}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x)=2x+6$ in key features of quadratic functions, evaluate $f(2)$.

a. 10 ✓ **b.** 12

c. 4 **d.** 14

Substitute $x=2$: $f(2)=2(2)+6=10$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x)=3x+7$ in solving quadratic equations by factoring, evaluate $f(2)$.

a. 13 ✓ **b.** 16

c. 6 **d.** 17

Substitute $x=2$: $f(2)=3(2)+7=13$.

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 8$ in solving quadratic equations using square roots, evaluate $f(2)$.

a. 16 ✓

b. 20

c. 8

d. 20

Substitute $x = 2$: $f(2) = 4(2) + 8 = 16$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for key features of quadratic functions consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5 ✓

d. 26.5

The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for solving quadratic equations by factoring consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5 ✓

d. 18.5

The sum is $10 + 14 + 18 + 24 = 66$, and dividing by 4 gives 16.5.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for solve quadratic equations using the quadratic formula consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5 ✓

d. 20.5

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)
For the model $f(x) = 5x + 9$ in solve quadratic equations using the quadratic formula, evaluate $f(4)$.

a. 29 ✓
b. 34
c. 20
d. 33

Substitute $x = 4$: $f(4) = 5(4) + 9 = 29$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)
A data set recorded for graphing quadratic functions consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5
b. 18.5
c. 19.5 ✓
d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)
For the model $f(x) = 6x + 3$ in graphing quadratic functions, evaluate $f(4)$.

a. 27 ✓
b. 33
c. 24
d. 31

Substitute $x = 4$: $f(4) = 6(4) + 3 = 27$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)
For the model $f(x) = 2x + 4$ in transformations of quadratics, evaluate $f(3)$.

a. 10 ✓
b. 12
c. 6
d. 14

Substitute $x = 3$: $f(3) = 2(3) + 4 = 10$.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the sum of all distinct real solutions to the equation $\frac{x}{x^2+4}+\frac{x}{x^2-3x+4}=\frac{1}{2}$?

a. 6 ✓

b. 7

c. 3

d. $3+\sqrt{5}$

To solve the equation, we first test if $x=0$ is a solution: $\frac{0}{0^2+4}+\frac{0}{0^2-3(0)+4}=0eq\frac{1}{2}$ Thus, $xeq0$. Since $xeq0$, we divide the numerator and denominator of each rational expression by x: $\frac{1}{x+\frac{4}{x}}+\frac{1}{x-3+\frac{4}{x}}=\frac{1}{2}$ Notice that the common algebraic structure $u=x+\frac{4}{x}$ appears in both denominators. Substituting u yields: $\frac{1}{u}+\frac{1}{u-3}=\frac{1}{2}$ Multiplying both sides by the common denomi.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Which of the following is equivalent to the expression $\frac{5}{x}+\frac{2}{3x}$ for all $xeq0$?

a. $\frac{17}{3x}$ ✓

b. $\frac{7}{4x}$

c. $\frac{7}{3x}$

d. $\frac{10}{3x^2}$

To add the rational expressions, first find a common denominator. The least common denominator of x and $3x$ is $3x$. Rewrite $\frac{5}{x}$ with the denominator $3x$ by multiplying both the numerator and denominator by 3: $\frac{5\cdot3}{x\cdot3}=\frac{15}{3x}$ Now add the two expressions with the common denominator: $\frac{15}{3x}+\frac{2}{3x}=\frac{15+2}{3x}=\frac{17}{3x}$ Common Misconceptions: - Choice B ($\frac{7}{4x}$) results from incorrectly adding both the numerators and denominators direct.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Which of the following expressions is equivalent to $\frac{x^2-4}{x-2}$ for all values of x where the expression is defined?

a. $x-2$

b. $x+2$ ✓

c. x

d. 2

The expression $\frac{x^2-4}{x-2}$ can be simplified by factoring the numerator. The numerator x^2-4 is a difference of squares, which factors as $(x-2)(x+2)$. Therefore, the expression becomes $\frac{(x-2)(x+2)}{x-2}$. For all values of x where $xeq2$ (so that the denominator is not zero), we can cancel out the common factor $(x-2)$ from the numerator and the denominator. This leaves us with $x+2$.

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Which rational expression is equivalent to $\frac{2x+4}{x+2}$?

a. $2x$

b. $x+2$

c. 2 ✓

d. $\frac{2}{x}$

Applying Rational Expressions & Equations principles, $\frac{2(x+2)}{x+2} = 2\ (x \neq -2)$. Therefore, the verified correct option is 2.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Simplify $\frac{x^2-25}{x-5}$, then evaluate the result at $x = 10$.

a. 14

b. 10

c. 17

d. **15** ✓

$\frac{x^2-25}{x-5} = x+5$ (for $x \neq 5$), so at $x = 10$: $10+5 = 15$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What value makes $\frac{x}{x-3}$ undefined?

a. 3 ✓

b. 0

c. -3

d. 1

Denominator $= 0$ when $x - 3 = 0 \Rightarrow x = 3$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Which of the following expressions is equivalent to $\frac{(x^4)^3}{x^5}$ for all $x \neq 0$?

a. x^2

b. x^7 ✓

c. $x^{\frac{12}{5}}$

d. x^{17}

To simplify the expression $\frac{(x^4)^3}{x^5}$, apply the foundational laws of exponents step-by-step: 1. **Power of a Power Rule**, $(x^a)^b = x^{a \cdot b}$: $(x^4)^3 = x^{4 \cdot 3} = x^{12}$ 2. **Quotient Rule**, $\frac{x^a}{x^b} = x^{a-b}$: $\frac{x^{12}}{x^5} = x^{12-5} = x^7$ **Common Misconceptions:** - Adding the exponents in the numerator instead of multiplying them gives $(x^4)^3 = x^{4+3} = x^7$, leading to $\frac{x^7}{x^5} = x^2$ (Choice A). - Dividing the exponents rather than subtracting them i.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Factoring Polynomials

Which of the following expressions is equivalent to $\sqrt{25x^4}$?

a. $\sqrt{5}x^2$

b. $5x^2$ ✓

c. $5x$

d. $25x^2$

We can simplify the expression $\sqrt{25x^4}$ by using the properties of radicals and exponents. Recall that $\sqrt{a \cdot b} = \sqrt{a} \cdot \sqrt{b}$ and $\sqrt{a^n} = a^{n/2}$. So, $\sqrt{25x^4} = \sqrt{25} \cdot \sqrt{x^4}$. We know that $\sqrt{25} = 5$ and $\sqrt{x^4} = x^{4/2} = x^2$. Therefore, $\sqrt{25x^4} = 5x^2$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 9.01 Radicals & Exponents

Rationalize the denominator: $\frac{8}{\sqrt{2}}$.

a. $\frac{8\sqrt{2}}{3}$

b. $8\sqrt{2}$

c. $\frac{8\sqrt{2}}{2}$ ✓

d. $\frac{9\sqrt{2}}{2}$

Multiply top and bottom by $\sqrt{2}$: $\frac{8}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{8\sqrt{2}}{2}$.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Linear Equations & Lines

Write $x^{2/3}$ using a radical.

a. $\sqrt{x^{3/2}}$

b. $\sqrt[2]{x^3}$

c. $(\sqrt{x})^3$

d. $\sqrt[3]{x^2}$ ✓

$x^{m/n} = \sqrt[n]{x^m} \Rightarrow x^{2/3} = \sqrt[3]{x^2}$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 11.01 Statistics & Data Analysis

Which is equivalent to $x^{-1/2}$?

a. $\sqrt{x}$

b. $\frac{1}{\sqrt{x}}$ ✓

c. $\frac{1}{x^2}$

d. $-\sqrt{x}$

$x^{-1/2} = \frac{1}{x^{1/2}} = \frac{1}{\sqrt{x}}$.

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Quadratic Functions

The graph of the exponential function $f(x) = a \cdot b^x$, where a and b are positive constants, is shown in the xy-plane below. The graph passes through the points $(-1, 8)$ and $(2, 1)$. If $f(k) = \frac{1}{16}$, what is the value of k?

a. **6 ✓**

b. 5

c. 4

d. -4

To find the parameters a and b, use the two given points on the graph of $f(x) = a \cdot b^x$: 1. Substitute $(-1, 8)$: $f(-1) = a \cdot b^{-1} = \frac{a}{b} = 8 \Rightarrow a = 8b$ 2. Substitute $(2, 1)$: $f(2) = a \cdot b^2 = 1$ 3. Substitute $a = 8b$ into the second equation: $(8b) \cdot b^2 = 1 \Rightarrow 8b^3 = 1 \Rightarrow b^3 = \frac{1}{8}$ Since $b > 0$, taking the cube root yields $b = \frac{1}{2}$. 4. Solve for a: $a = 8\left(\frac{1}{2}\right) = 4$ Thus, the function is.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Exponents & Polynomials

A line of best fit for data has equation $y = 3x + 2$. Predict y when $x = 5$.

a. 20

b. 17 ✓

c. 15

d. 13

Applying Statistics & Data Analysis principles, $y = 3(5) + 2 = 17$. Therefore, the verified correct option is 17.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Functions & Transformations

A test consists of 25 multiple-choice questions. A student scored 80%. If the student correctly answered 4 more questions, what would the new percentage score be?

a. 84%

b. 88%

c. 92%

d. **96% ✓**

Initial correct: $0.80 \times 25 = 20$. New correct: $20 + 4 = 24$. New score: $24/25 = 96\%$.

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

A sequence of positive real numbers $(a_n)_{n \geq 1}$ satisfies $a_1 = 1$, $a_2 = 3$, and the recurrence relation $a_{n+1} = \frac{a_n^2 + 2}{a_{n-1}}$ for all integers $n \geq 2$. What is the value of the infinite series $\sum_{n=1}^{\infty} \frac{1}{a_n a_{n+1}}$?

a. $\frac{\sqrt{3}-1}{2}$ ✓

b. $\sqrt{3} - 1$

c. $\frac{3-\sqrt{3}}{2}$

d. $\frac{2-\sqrt{3}}{2}$

We solve the problem in three main stages: converting the non-linear recurrence into a linear recurrence, establishing a telescoping identity for the series, and computing the limiting ratio of consecutive terms. **Step 1: Linearizing the recurrence relation** From the given recurrence, clearing denominators gives: $a_{n+1}a_{n-1} - a_n^2 = 2$ for all $n \geq 2$. Shifting the index by 1 gives: $a_{n+2}a_n - a_{n+1}^2 = 2$. Equating both expressions yields.

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